BEE MIDSEM II (ROB)

MM—15 MT—50min

Q1. (a) Define Slip in 3 phase induction motor and explain why induction motor cannot run at synchronous speed. (2)

(b) Draw Torque –Slip characterstic of 3 phase induction motor.(1)

(c) A 3 phase 6-pole induction motor is connected to a 60 hz supply. The voltage induced in the rotor bar is 4 V when the motor is at stand still. Calculate the voltage and frequency induced in the rotor bar at 300 r.p.m. (2)

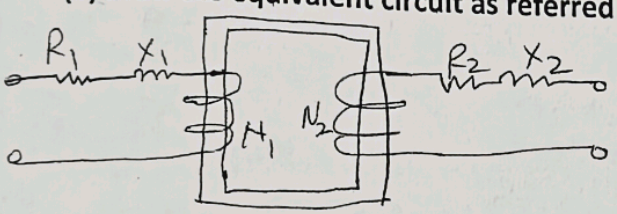
Q2. (a) Derive the EMF equation of DC Generator. (2)

(b) Write the DC Motor principle and what is the significance of Back emf in DC Motor. (2)

(c) Convert $(17.625)_{10}$ to hexadecimal number. (1)

Q3. (a) Define voltage transformation ratio in transformer. (1)

(b) Draw the equivalent circuit as referred to primary for the following. (1)



(c) A 4 pole lap wound dc shunt generator has a useful flux per pole as 0.07 wb . The armature winding consists of 220 turns , each of 0.004 ohm ressistance .Calculate the terminal voltage when running at 900 r.p.m if the armature current is 50 A . (3)